Crypto Mean Reversion Research Report (Binance USDT-M Perpetual)

Prepared 16 October 2026. Research data period: December 2019 to December 2023 (train). OOS data (2024 onward) has never been touched.

1. Executive summary

This research tests whether a z-score mean reversion strategy on 1-hour candles in Binance perpetuals has an edge that survives costs and realistic execution. Conclusions:

- **The original edge was illusory.** The original strategy (price z-score, z ≥ 3.8, TP 2.5 ATR, SL 0.5 ATR placed only after the 4th bar) looked excellent in the old backtest: +0.463% per trade, PF 1.84, 5 of 5 years positive. That backtest filled stops at the stop price. In real execution, a position that has already passed the SL by the time the SL may be placed is closed at the market price. With a model that mimics the live code, the result is -0.135% per trade, PF 0.89, 0 of 5 years positive.
- **All execution fixes failed.** Entry timing, candle confirmation, 15m order flow features, wider SL/TP, time stops, and catastrophic stops did not restore the edge.
- **One new source of information helped:** computing the z-score on returns residualized against BTC, plus a market breadth filter. The derived candidate (long-only, window 100, threshold 4.0, 3-hour hold, no SL) looked viable in train, but most of its edge came from pairs selected using the same train data (selection bias).
- **The clean test failed the candidate.** On 202 pairs that were never selected, net per trade in validation was only +0.063% (fee 0.12%) and -0.017% (fee 0.20%), with a validation ret/DD of 0.14 at fee 0.20%. Past pair performance also did not predict future performance (mean IC -0.037, t -0.97).
- **Decision:** the strategy was discontinued, the OOS was not used, and the demo bot running the old strategy should be shut down. A possible next direction is market-neutral funding carry, which has not yet been tested.

2. Original strategy and infrastructure

Signal (LIVE/demo version and train backtest):

- z-score of close, window 50, entry when |z| ≥ 3.8 (long when z is negative, short when z is positive).
- Filters: ATR percentile ≤ 0.85 (ATR 14, lookback 100) and half-life ≤ 20 bars (window 100). ADX not used.
- Entry at the open of the bar after the signal, one position per pair, round-trip fee 0.09%.

Exit: native TP of 2.5 ATR active from bar 1. Native SL of 0.5 ATR placed only after the 4th bar closes (`MIN_HOLD_BARS = 3`). Live trading adds a 30% catastrophic stop. If the price has already passed the SL level when the SL is placed (error -2021), the position is closed at market.

Initial universe and sizing: 50 pairs in `pairs_train_selected.txt`, maximum 2 positions, margin tier 0.5x, leverage 1x. The pair list was chosen by profit factor above 1 on train data (using the old strategy).

3. Research protocol

- All analysis uses data up to 2023-12-31. Train is split into discovery (before 2022-07-01) and validation (after). OOS is kept clean.
- Pass criteria were written before results were seen. Distress fees range from 0.09% to 0.25%.

- Execution is modeled realistically: late SLs and gaps are closed at market price, not at the stop price.
- Point-in-time universe: liquidity and age filters use only information available before the bar, and delisted coins are kept (survivorship-free).
- t-stats are computed per day (clustered), because trades across pairs are correlated. For large grids, the best t-stat expected from noise is about sqrt(2 ln N): roughly 3.5 for 450 configurations.

4. Stage 1: audit of the old backtest

Train trades: 1535 (50 pairs). Four exit models on the same trades:

Model	n	Win %	Avg return/trade	PF	Positive years
ideal, SL delayed 3 bars (old backtest)	1535	29.0	+0.463%	1.84	5/5
live, SL delayed 3 bars (realistic SL gap)	1532	30.4	-0.135%	0.89	0/5
ideal, no grace period (SL from bar 1)	1637	14.5	-0.114%	0.83	0/5
live, no grace period	1637	14.5	-0.114%	0.83	0/5

Cause: 31% of trades ended as SLGAP, meaning the price had already passed the SL when the SL could be placed, so the position was closed at market far from the SL level. Example from the demo: a LUNA2 trade with a planned SL of -0.4% realized -4.69%. In addition, 69.4% of trades hit an adverse excursion of more than 0.5 ATR within the first 3 bars, so delaying the SL is strictly necessary for this strategy, and that delay is what creates the hidden loss.

5. Stage 2: attempts to fix timing and exit

- **Signal candle features (13 features):** AUC against early SL was only 0.43 to 0.54. A reversing-z confirmation filter could not be tested because only 1 of 1535 trades satisfied it (signals always appear on a candle still moving against the position).
- **Delayed entry and wider SL/TP (160 configurations, SL active from bar 1):** all negative or zero. Best: -0.099% per trade, 1 of 5 years positive. Widening the SL made results worse.
- **TP + catastrophic stop + time stop (80 configurations):** best +0.011% (PF 1.01, t 0.14). The pattern is small wins, large losses: win rate around 80% but PF below 1, with worst trades of -20% to -92%.

Conclusion of this stage: the failure is not in entry timing; the 1h price signal itself gives no edge after costs.

6. Stage 3: BTC-residual z-score and regime filter

Hypothesis: many falling knives are market moves, not coin-specific deviations. Z is computed on the cumulative coin return minus beta (rolling 500 bars, no lookahead) times the BTC return.

- Across 64 configurations, residual mode was always better than raw price z. Residual gross was positive (+0.07% to +0.19%) while raw z was negative.
- A breadth filter (number of pairs in the same extreme direction on the same bar ≤ 5) raised the average from +0.024% to +0.143%. A BTC volatility filter hurt (-0.059%).
- Candidate: residual, z 3.8, breadth, live exit (SL gap-aware): n 1486, +0.217% per trade (gross +0.307%), PF 1.19, 3 of 4 years positive.

Candidate robustness (train):

- 144-cell grid (beta window, breadth, threshold, exit): 100% positive.
- Daily bootstrap: 95% CI of the mean [+0.009%, +0.420%]; cluster t 0.76. The edge is not statistically proven.
- Long is much stronger than short: long n 460, +0.569%, PF 1.79; short n 1026, +0.059%, PF 1.05.

- Concentration: top 20 trades = 57.5% of total return, top 3 months = 77.7%. Without the top 10 pairs, the average falls to +0.016%.
- Positions are correlated: 1994 overlapping trade pairs, 92.4% in the same direction.
- Sizing changes the picture: 2 slots x 50% gives +19.5% return with max DD -58% (ret/DD 0.33). 5 slots x 20% gives +44.8% with DD -26% (ret/DD 1.69). 10 slots x 10% gives +27.5% with DD -13% (ret/DD 2.07). Diversification improves the ratio but does not create an edge.

7. Stage 4: 15m order flow, funding, and open interest

Data was downloaded from Binance Vision: 15m klines (taker buy volume for delta), metrics (open interest), and funding. Availability notes: open interest for newer altcoins is only available from 2021-12-01 (BTC from 2020-09-01), liquidation data is no longer published by Binance, and tick-level CVD is too heavy for 50 pairs.

Diagnostics of 15 features against candidate trade outcomes (TP vs SLGAP): all AUCs fall between 0.46 and 0.55. The feature with the largest edge (`fund_orient`, 0.051) is non-monotonic and unstable. There was a bug: open interest was computed from USD value, which moves with price; after fixing it to contract quantity, OI AUC was 0.49 to 0.50. Conclusion: no additional signal from this data.

8. Stage 5: screening 450 ideas with the same engine

Engine: entry at the open after the signal, fixed hold, fee 0.12%, one position per pair, no SL. Idea families: cross-sectional reversal/momentum (BTC residual), per-coin residual z, Donchian breakout, and extreme funding; timeframes 1h, 4h, 1d.

- Median of each idea family: gross 0% and net about -0.12%, exactly the size of the fee. Most ideas have no edge.
- 1d trend breakout dominated discovery (net 3% to 7% per trade) but was negative in validation. That is a 2020-2021 bull regime effect, not an edge that holds in all conditions.
- Only 2 of 109 shortlisted ideas passed the validation criteria. Both are in the residual z family, reversal mode, window 100, threshold 3.5, hold 3 or 6 hours, so they are really one finding.

9. Stage 6: survivor (residual z, no SL)

Execution is simple: enter at the open after the signal, exit at the open after H bars, no matter what. No TP or SL.

- **Threshold 3.5, both sides:** n 5521, +0.249% per trade, PF 1.17. Discovery +0.377%, validation +0.116%. Worst trade -71.9% (short SFP, October 2021), p1 -17.6%. Parameter neighbors: 51% of combinations positive in both periods at fee 0.12%, only 19% at fee 0.20%.
- **Catastrophic stop makes it worse** at every level (a 20% stop lowers discovery net from 0.38% to 0.08%). Tail risk must be controlled through position sizing.
- **Threshold 4.0:** n 2880, +0.471% per trade (discovery +0.733%, validation +0.220%), PF 1.31, validation net stays positive up to fee 0.25%. Neighbors: 65% positive at fee 0.12%, 38% at fee 0.20%. Nine of the ten worst trades are shorts (including three correlated trades on 13 July 2023).
- **Long-only, threshold 4.0:** n 1078, +1.056% (discovery +1.967%, validation +0.282%), PF 1.93 (discovery 2.17, validation 1.42). Worst trade -26%, p1 -16%. Validation net +0.20% at fee 0.20%. Portfolio H=3, 5 slots, 20% per position: 27.6% return per year with max DD -12.3%; validation 16.2% per year with DD -9.1% (ret/DD 2.68; 1.79 at fee 0.20%). This candidate was locked for a single OOS run but has not been run.

10. Stage 7: broad universe, selection bias, and pair persistence

Broad universe. The symbol list was taken from the Binance Vision bucket listing (including delisted symbols), with point-in-time volume and age filters. The long-only threshold 4.0 candidate was tested under three universe rules:

Rule	n	Net/trade	PF	Validation ret/DD at fee 0.20%
daily volume ≥ 5M USD	2416	0.530%	1.40	0.67
daily volume ≥ 20M USD	1606	0.894%	1.61	1.95
top 60 by volume	1057	1.013%	1.77	1.83

Selected pairs vs never-selected pairs (validation net at fee 0.12%):

Rule	Other pairs	Selected pairs
volume ≥ 5M	+0.063% (138 pairs)	+0.320% (48 pairs)
volume ≥ 20M	+0.135% (113 pairs)	+0.375% (48 pairs)
top 60	-0.003% (84 pairs)	+0.384% (43 pairs)

The advantage of the selected pairs is largely selection bias: the list was built from profit factor above 1 using the old strategy on the same train data (including the validation period), with a backtest that was still too optimistic at the time.

Pair persistence test (walk-forward). Pair quality was measured with a dense signal (threshold 2.5), 12-month lookback, 6-month test period.

- Cross-period IC: mean -0.037, t -0.97, positive in only 2 of 6 periods.
- Top vs bottom tercile on survivor trades: difference +0.34%, t 0.64 (not significant). On the dense signal: -0.03%, t -0.54.
- Structural features with PnL in the theoretical direction but weak: lag-1 autocorrelation of residual returns (IC -0.079, negative in 6 of 6 periods) and variance ratio (IC -0.056). Their t-values tend to be overstated because the lookback windows overlap.

Conclusion: past pair performance does not predict future performance, so PnL-based pair selection is noise.

11. Final test: universe without selected pairs

Conditions: 202 pairs outside `pairs_train_selected.txt`, volume ≥ 5M USD, age ≥ 180 days, long-only, threshold 4.0, window 100, no SL.

- Per trade: n 1476, +0.317% (discovery +0.790%, validation +0.063%), PF 1.22 (validation 1.07). At fee 0.20%, validation net is -0.017%.
- Best portfolio (H=3, 10 slots): 3.4% return per year with max DD -25%, ret/DD 0.54 (full). Validation 3.7% per year, ret/DD 0.80.
- Fee stress at 8 slots: validation ret/DD 0.63 (fee 0.12%), 0.14 (0.20%), -0.12 (0.25%).
- Tail losses remain large: FTT -51.9% (November 2022) and a single day, 11 May 2022, that dragged down a dozen pairs at once.

Assessment against the criteria written before the test: validation net at fee 0.20% must be positive (failed, -0.017%) and validation ret/DD at fee 0.20% must be at least 1 (failed, 0.14). **Result: failed.** The OOS was not run.

12. Lessons

1. **The execution model determines the result.** A backtest must mimic live code behavior, including late stops and gaps. The change from +0.463% to -0.135% came entirely from fill assumptions.
2. **Selecting pairs from the same data is data snooping.** Selected pairs look good in the validation that was also used to select them. A persistence test and a group of never-selected pairs are needed.

3. **Many configurations, much noise.** Of 450 ideas, 2 passed validation and both were the same finding. The best result from many tries must be compared with the t-stat expected from noise.
4. **Edge disappears across regimes.** Net per trade fell from discovery (2020-2022) to validation (2022-2023) in almost all variants.
5. **Tail events hit many pairs on the same day** (12 March 2020, 11 May 2022, 13 July 2023). Tight stops do not help; small position sizes and many slots improve the return/DD ratio without creating an edge.
6. **Pure price signals on 1h in liquid Binance perpetuals are heavily crowded.** The remaining edge after fees and slippage is thin or zero.

13. Status and recommendations

- Shut down the demo bot for the old strategy (z 3.8, delayed SL). In realistic simulation, that strategy loses money.
- Do not spend the OOS on the long-only candidate. The parameters that were locked (not run): residual z window 100, long-only when z ≤ -4.0, 3-hour hold, 5 slots, 20% per position, no SL.
- This research protocol can be reused for other ideas: point-in-time universe, criteria first, discovery/validation, persistence test, and a single OOS run.

14. Appendix: scripts and limitations

Scripts (chronological): `generate_train_trades_locked_2.py`, `analyze_trades.py`, `diagnose_entry_features.py`, `sweep_entry_exit.py`, `sweep_time_stop.py`, `sweep_residual.py`, `robust_residual.py`, `diagnose_orderflow.py`, `download_binance_vision.py`, `screen_ideas.py`, `robust_survivor.py`, `oos_survivor.py`, `build_universe.py`, `pair_persistence.py`.

Limitations:

- All numbers come from train data. Parameters such as threshold and number of slots were chosen while looking at the validation period, so validation is not fully clean data.
- Fees of 0.09% to 0.25% are assumptions, not slippage measured from demo fills. Measuring the gap between fill price and bar open price from the demo log is still an open step.
- Portfolio returns are non-compounding and drawdown is computed from hourly closes; intra-hour drawdown is not captured.
- Binance Vision 1h data is used as-is; some pairs have short histories or stop because of delisting.